

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I **C. PRIYANKA (192511074)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. PRIYANKA

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- ii. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- iii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

ANSWER:

Criteria 1: Understanding of Case

Problem Statement

A hospital has developed a **Rule-Based Logical Agent** to perform preliminary diagnosis based on patient symptoms. The system stores medical knowledge as logical rules and patient symptoms as facts. The objective is to infer possible diseases using logical reasoning.

Knowledge Base

- | | |
|--|----|
| Rule | 1: |
| If Fever AND Cough, then Possible Flu | |
| Rule | 2: |
| If Fever AND Rash, then Possible Measles | |
| Rule | 3: |
| If Cough AND Breathlessness, then Possible Pneumonia | |

Facts

Patient A:

- Fever = True
- Cough = True
- Breathlessness = True

Criteria 2: Application of Concepts

(a) Propositional Logic Representation

Symbols

Symbol	Meaning
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F	Fever
C	Cough
R	Rash
B	Breathlessness

Symbol Meaning

Flu Possible Flu

Mea Possible Measles

Pne Possible Pneumonia

Rules

Rule 1

Rule 2

Rule 3

Facts**(b) Forward Chaining**

Initially Known Facts

F

C

B

Step 1

Check Rule 1

$F \wedge C \rightarrow \text{Flu}$

Both conditions are True.

Rule Fired

New Fact Derived

Flu

Step 2

Check Rule 2

$F \wedge R \rightarrow \text{Mea}$

Rash is absent.

Rule cannot fire.

Step 3

Check Rule 3

$C \wedge B \rightarrow \text{Pne}$

Both conditions are True.

Rule Fired

New Fact Derived

Pne

Final Facts

F

C

B

Flu

Pne

Final Diagnoses

- Possible Flu
- Possible Pneumonia

(c) Backward Chaining

Goal

Pne

Search for rule producing Pne.

Found

$C \wedge B \rightarrow \text{Pne}$

Need to prove

C

and

B

Knowledge Base contains

C = True

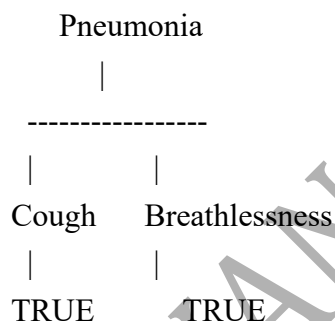
B = True

Both subgoals are satisfied.

Hence

Pne = True

Goal Reduction Tree



Goal Verified Successfully.

Criteria 3: Analysis

Forward Chaining begins from known facts and derives every possible diagnosis.

Backward Chaining begins with the desired diagnosis and verifies whether the required symptoms exist.

Since Patient A has both cough and breathlessness, the logical agent correctly confirms **Possible Pneumonia**. The system also detects **Possible Flu** because fever and cough are present. No conclusion about measles can be drawn because rash is absent.

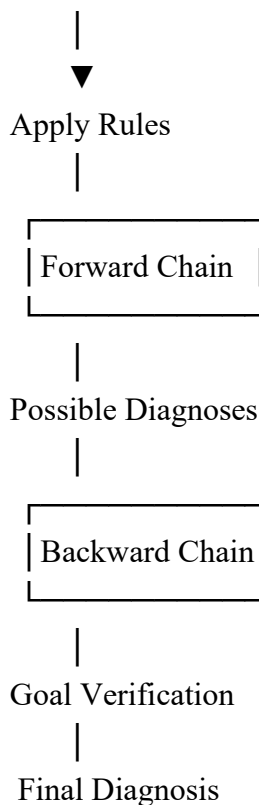
Criteria 4: Solution Design

Logical Diagnosis Process

Patient Symptoms



Knowledge Base



Criteria 5: Clarity

Result

The rule-based logical agent correctly diagnosed Patient A with:

- Possible Flu
- Possible Pneumonia

The backward chaining process successfully verified the diagnosis of Pneumonia. Therefore, the logical reasoning process is correct and consistent.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: Soil Dry $\rightarrow$ Irrigation Needed
- 2) P2: Irrigation Needed $\wedge$ Crop Wheat $\rightarrow$ Apply Drip Method
- 3) P3: $\neg$ Apply Drip Method $\wedge$ Crop Wheat $\rightarrow$ Crop At Risk
- 4) Facts: Soil Dry = True, Crop Wheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'Apply Drip Method'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'Soil Dry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'Crop At Risk' still follow? Apply appropriate reasoning patterns to justify.

ANSWER:

Criteria 1: Understanding of Case

Problem Statement

An agricultural expert system helps farmers decide irrigation methods using propositional logic. The system stores farming knowledge as logical rules and crop conditions as facts. The objective is to determine whether drip irrigation should be applied and analyze crop risk.

Knowledge Base

P1

Soil Dry $\rightarrow$ Irrigation Needed

P2

Irrigation Needed $\wedge$ Crop Wheat $\rightarrow$ Apply Drip Method

P3

$\neg$ Apply Drip Method $\wedge$ Crop Wheat $\rightarrow$ Crop At Risk

Facts

Soil Dry

Crop Wheat

Criteria 2: Application of Concepts

(a) CNF Conversion

P1

Original

Soil Dry $\rightarrow$ Irrigation Needed

Remove implication

$\neg$ Soil Dry $\vee$ Irrigation Needed

Already CNF.

P2

Original

Irrigation Needed $\wedge$ Crop Wheat $\rightarrow$ Apply Drip Method

Remove implication

$\neg$ (Irrigation Needed $\wedge$ Crop Wheat) $\vee$ Apply Drip Method

Apply De Morgan's Law

$\neg$ Irrigation Needed $\vee \neg$ Crop Wheat $\vee$ Apply Drip Method

CNF obtained.

P3

Original

$\neg$ Apply Drip Method $\wedge$ Crop Wheat $\rightarrow$ Crop At Risk

Remove implication

$\neg$ ($\neg$ Apply Drip Method $\wedge$ Crop Wheat) $\vee$ Crop At Risk

Apply De Morgan's Law

Apply Drip Method $\vee \neg$ Crop Wheat $\vee$ Crop At Risk

CNF obtained.

Facts

Soil Dry

Crop Wheat

Final CNF Clauses

C1

$\neg$ Soil Dry $\vee$ Irrigation Needed

C2

$\neg$ Irrigation Needed $\vee$ $\neg$ Crop Wheat $\vee$ Apply Drip Method

C3

Apply Drip Method $\vee$ $\neg$ Crop Wheat $\vee$ Crop At Risk

C4

Soil Dry

C5

Crop Wheat

(b) Resolution Refutation

Goal

Apply Drip Method

Negated Goal

$\neg$ Apply Drip Method

Add to KB.

Resolution Steps

Resolve

$\neg$ Soil Dry $\vee$ Irrigation Needed

with

Soil Dry

Result

Irrigation Needed

Resolve

$\neg$ Irrigation Needed $\vee$ $\neg$ Crop Wheat $\vee$ Apply Drip Method

with

Irrigation Needed

Result

$\neg$ Crop Wheat $\vee$ Apply Drip Method

Resolve

$\neg$ Crop Wheat $\vee$ Apply Drip Method

with

Crop Wheat

Result

Apply Drip Method

Resolve

Apply Drip Method

with

$\neg$ Apply Drip Method

Result

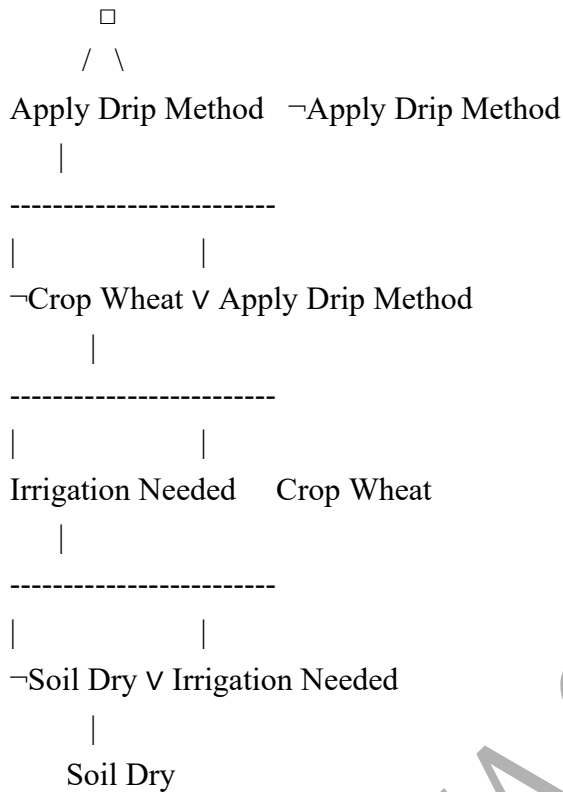
□

(Empty Clause)

Hence,

Apply Drip Method is proved.

Resolution Proof Tree



(c) Analysis if Soil Dry is Removed

Remaining Facts

Crop Wheat

Only.

Without Soil Dry,

Irrigation Needed

cannot be derived.

Without Irrigation Needed,

Apply Drip Method

cannot be proved.

Without proving

Apply Drip Method

there is no logical basis to infer

Crop At Risk

because the rule requires the explicit condition $\neg$ **ApplyDripMethod**, which is not established merely by failing to prove Apply Drip Method.

Therefore, the expert system cannot conclude either **Apply Drip Method** or **Crop At Risk** from the remaining knowledge.

Criteria 3: Analysis

The CNF conversion simplifies all rules into a standard form suitable for automated theorem proving. Resolution refutation successfully proves that **Apply Drip Method** is true by deriving the empty clause. Removing the fact **Soil Dry** breaks the inference chain, preventing the derivation of **Irrigation Needed** and consequently **Apply Drip Method**. This demonstrates the importance of complete and accurate facts in knowledge-based reasoning.

Criteria 4: Solution Design

Logical Inference Process

Input Facts



Convert Rules to CNF



Resolution Engine



Apply Resolution Steps



Derive Apply Drip Method



Empty Clause ($\square$)



Goal Successfully Proved

Criteria 5: Clarity

Result

The agricultural expert system correctly proves **Apply Drip Method** using CNF conversion and resolution refutation. When the fact **Soil Dry** is removed, the logical inference chain is interrupted, so neither **Apply Drip Method** nor **Crop at Risk** can be concluded. The reasoning process is systematic, logically justified, and clearly demonstrates the operation of a propositional logic-based expert system.

PRIYANKA.C-191511074